

Multiple Choice (10 marks)

1. What pressure would 131g of xenon gas in a vessel of volume 1.0dm^3 exert at 25°C if it behaved as a van der Waals gas?
 - (a) 22atm
 - (b) 20atm
 - (c) 15atm
 - (d) 30atm
 - (e) 34atm
2. Determine the equilibrium constant for the reaction $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ the equilibrium concentrations are: H_2 , 0.9 moles/liter; I_2 , 0.4 mole/liter; HI, 0.6 mole/liter.
 - (a) 0.9
 - (b) 1.0
 - (c) 2.5
 - (d) 0.3
 - (e) 0.5
3. The temperature of the fireball in a thermonuclear explosion can reach temperatures of approximately 10^7 K. What value of λ_{max} does this correspond to?
 - (a) 2×10^{-10} m
 - (b) 1×10^{-10} m
 - (c) 7×10^{-10} m
 - (d) 6×10^{-10} m
 - (e) 4×10^{-10} m
4. The density of a gaseous compound was found to be 1.23kgm^{-3} at 330K and 20kPa. What is the molar mass of the compound?
 - (a) 169 gmol^{-1}
 - (b) 123 gmol^{-1}
 - (c) 200 gmol^{-1}
 - (d) 150 gmol^{-1}
 - (e) 190 gmol^{-1}
5. A liquid element that is a dark-colored, nonconducting substance at room temperature is

- (a) lead
- (b) cesium
- (c) bromine
- (d) francium
- (e) rubidium

True / False (10 marks)

Answer True or False

6. True or False: The height of a mercury column corresponding to a 600 mm water column is approximately 44.1 mm Hg due to the density difference.
7. True or False: The positron is the most massive subatomic particle among the listed options.
8. True or False: The logarithm of 0.0364 is approximately -1.4389.
9. True or False: An electron transitioning from energy level 4 to 1 emits a photon with a wavelength of 1.03×10^{-7} m.
10. True or False: The combustion of 1000 cubic feet of butane gas requires 5500 cubic feet of oxygen for complete reaction.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the volume of concentrated sulfuric acid needed to achieve a mass of 74.0 g, given its density of 1.85 g/ml. Discuss the implications of density in this context.
12. A chemist dropped a 200 g object into water, displacing 60 ml, and later displaced 50 g of oil. Calculate and analyze the densities of both the object and the oil, discussing the implications of your findings.

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